LINEAR MODELS

Log Transformations

REGRESSION ANALYSIS

Elisa Vaca · ITESO · Otoño 2026

What we found in the credit model

1

What we did

Model 1 (simple) and Model 2 (multiple) predict the **credit line in pesos** from client characteristics.

2

What did not hold

Linearity. Both models fail the Ramsey RESET test, so a **straight line is the wrong shape for this relationship.**

3

Recommendation

Log transformations. Then run RESET again and compare the models.

A question a straight line cannot answer



Money works in percentages

Income and credit lines are positive and right-skewed. A 1,000-peso raise means a lot to a low earner and almost nothing to a high earner.

A straight line gives one peso answer to everyone. A log model gives a percentage answer.

THE RISK TEAM ASKS

If a client earns 10% more, how much more credit should we give?

That is a percentage question. A straight line in pesos answers a different one.

Three reasons to bend a variable

Linearity

A curved relationship becomes a straight line. This is the problem RESET detected.

Skewness and spread

Long tails and residuals that fan out get tamed. Outliers lose their grip.

Interpretation

Effects are told in percentages, the language of business.

Standardization and normalization only rescale a variable. They do not change its shape. Today we use transformations that do.

Four ways to bend a variable

Transformation	What it does	Use it when	Skew it fixes
ln(x)	Compresses large values strongly, expands small ones	Money, prices, growth: effects are multiplicative	Right skew (strong)
√x	Compresses large values gently	Counts, or mild long tails	Right skew (mild)
1 / x	Compresses large values and reverses the order	Rates, or very strong long tails	Right skew (very strong)
x^2	Expands large values	Relationships that accelerate	Left skew

Today's focus: ln(x). It is the only one that also gives a clean percentage reading.

Doubling always adds the same amount in logs

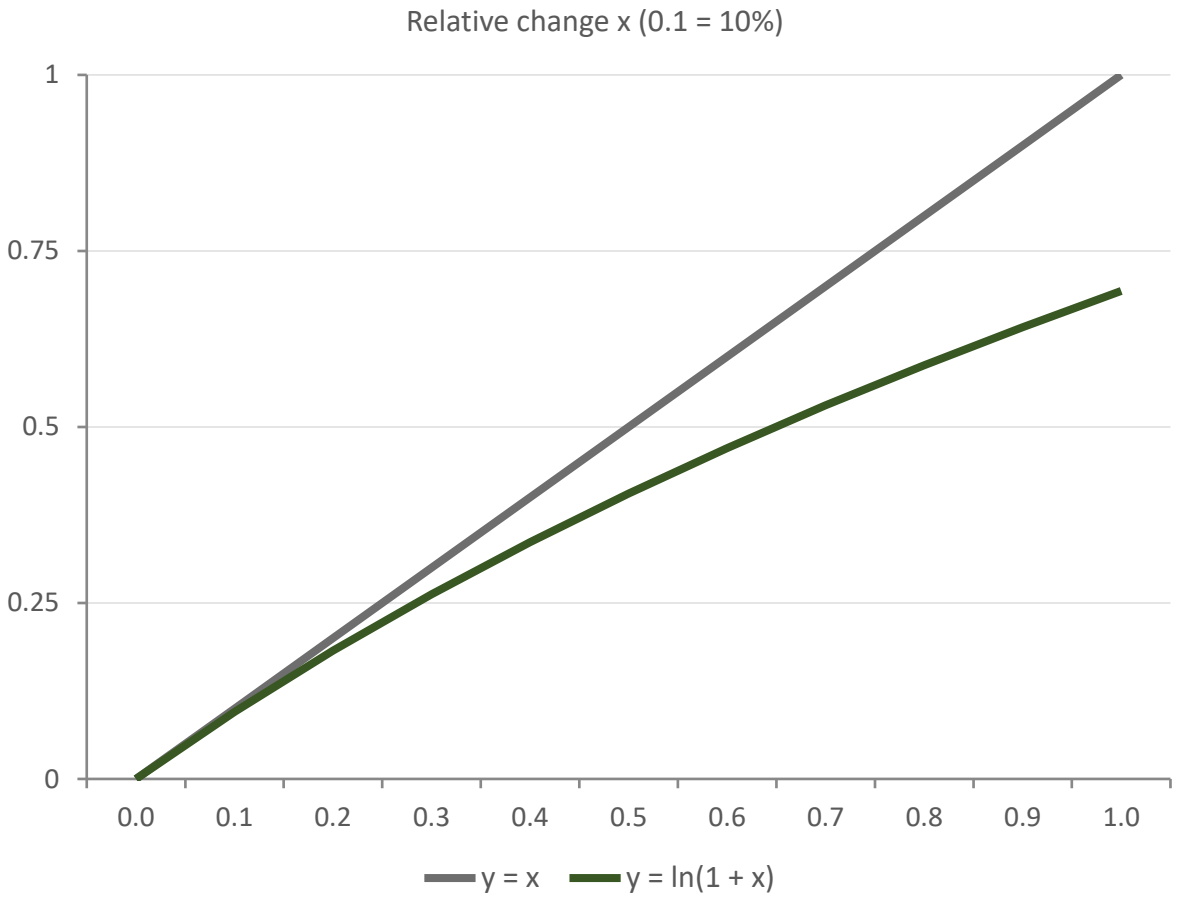
Monthly income (pesos)	5,000	× 2 →	10,000	× 2 →	20,000	× 2 →	40,000
			▼ *take ln*				
Natural log of income	8.52	+ 0.69 →	9.21	+ 0.69 →	9.90	+ 0.69 →	10.60

$$\ln(a \times b) = \ln(a) + \ln(b)$$

Multiplicative effects become additive. And a linear model is additive.

Not just money: any positive quantity that changes by ratios, like growth, sizes, prices or waiting times.

For small changes, ln(1 + x) ≈ x



Small changes: ln(1 + x) ≈ x

At 1% the gap is invisible. At 10% it is 0.095. At 50% it breaks: 0.405.

Only positive numbers

ln(x) needs x > 0. With zeros, ln(1 + x) is a common fix, but it changes the reading slightly.

Always use the natural log

The base does not change the fit, but only ln gives the clean "β ≈ percent" reading.

One question: where does the log go?

	X in original scale	X in log scale
Y in original scale	**LIN–LIN** $y = \alpha + \beta \cdot x + \varepsilon$ **Nobody is in %**	**LIN–LOG** $y = \alpha + \beta \cdot \ln(x) + \varepsilon$ **x is in %**
Y in log scale	**LOG–LIN** $\ln(y) = \alpha + \beta \cdot x + \varepsilon$ **y is in %**	**LOG–LOG** $\ln(y) = \alpha + \beta \cdot \ln(x) + \varepsilon$ **Both are in %**

LIN–LIN: the baseline you already know

LIN–LIN $y = \beta_0 + \beta_1 \cdot x + \varepsilon$

HOW TO READ β

$$\beta = \Delta y / \Delta x$$

For every one-unit increase in x, y changes by β units.

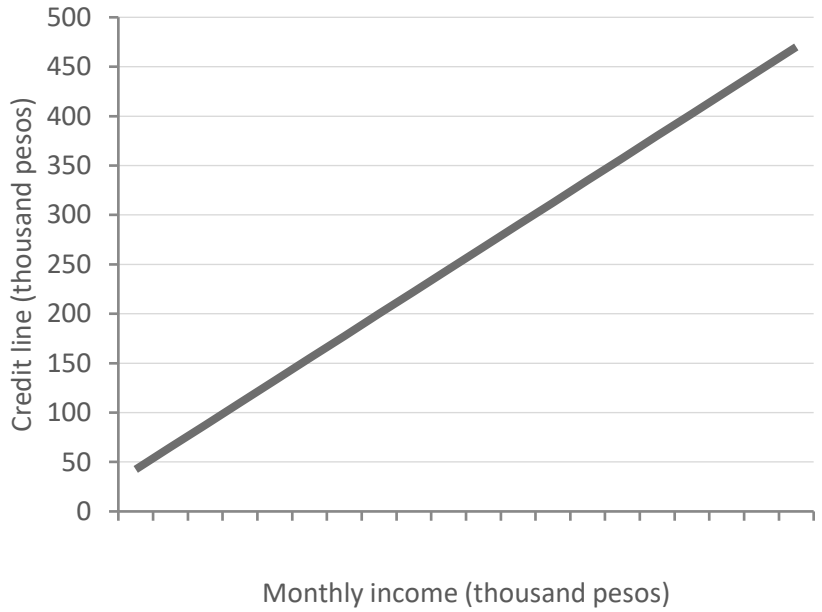
EXAMPLE (ILLUSTRATIVE VALUES)

β = 4.5: each extra 1,000 pesos of monthly income goes with 4,500 pesos more credit line.

WHERE IT COMES FROM

$y_1 - y_0 = \beta \cdot (x_1 - x_0) \rightarrow \Delta y = \beta \cdot \Delta x$

Shape: a straight line



The same slope for every client, rich or poor. This is the assumption RESET rejected.

LIN–LOG: log on x, effect in pesos per 1%

LIN–LOG $y = \beta_0 + \beta_1 \cdot \ln(x) + \varepsilon$

HOW TO READ β

$$\beta = \Delta y / \Delta \%x$$

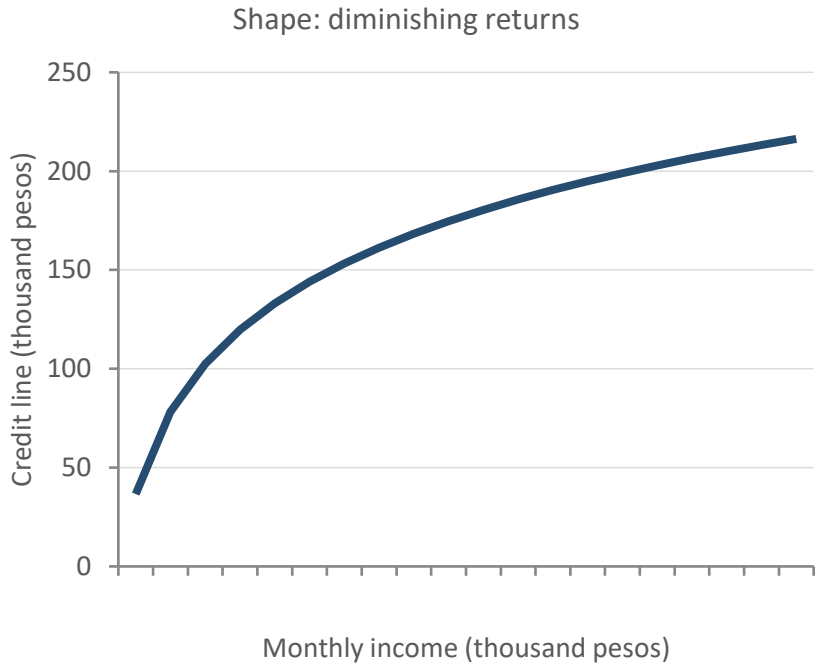
For every 1% increase in x, y changes by β / 100 units.

EXAMPLE (ILLUSTRATIVE VALUES)

β = 60 (thousand pesos): 1% more income goes with 0.6 thousand, or 600 pesos, more credit line.

WHERE IT COMES FROM

$\Delta y = \beta \cdot [\ln(1.01x) - \ln(x)] = \beta \cdot \ln(1.01) \approx \beta / 100$



Needs x > 0. Good when each extra unit of x adds less than the one before.

LOG–LIN: log on y, effect in % per unit

LOG–LIN $\ln(y) = \beta_0 + \beta_1 \cdot x + \varepsilon$

HOW TO READ β

$\beta = \Delta\%y \,/\, \Delta x$

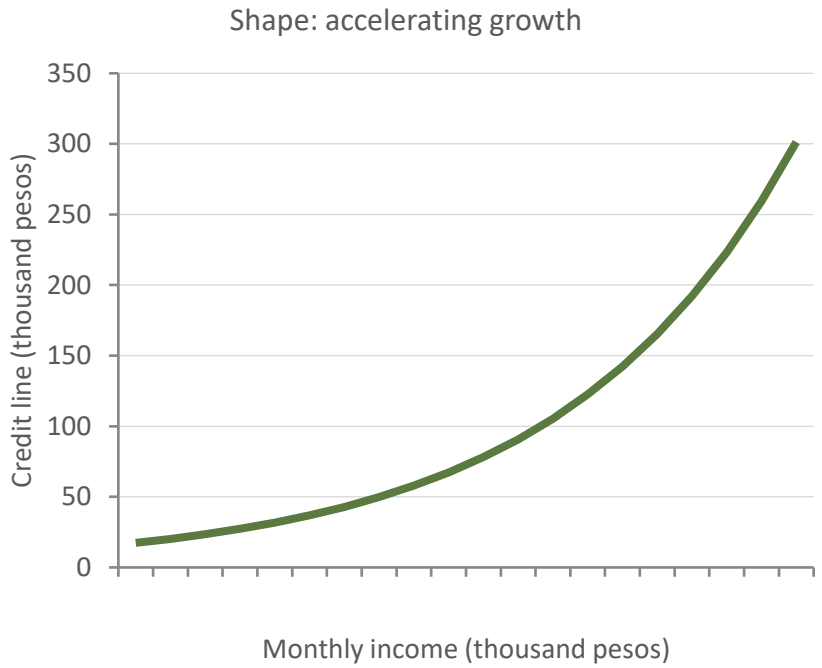
For every one-unit increase in x, y changes by about 100·β percent.

EXAMPLE (ILLUSTRATIVE VALUES)

β = 0.03: each extra 1,000 pesos of income goes with about +3% credit line (exact: e^0.03 − 1 = 3.05%).

WHERE IT COMES FROM

$\ln(y_1) - \ln(y_0) = \beta \;\rightarrow\; y_1 / y_0 = e^{\beta} \;\rightarrow\; \Delta\%y = e^{\beta} - 1 \approx \beta$



Needs y > 0. The assumption is on ln(y): errors are normal in log scale, so y is lognormal (not “y is Normal”).

LOG–LOG: elasticity, % per %

LOG–LOG $\ln(y) = \beta_0 + \beta_1 \cdot \ln(x) + \varepsilon$

HOW TO READ β

$\beta = \Delta\%y \,/\, \Delta\%x$

For every 1% increase in x, y changes by about β percent.

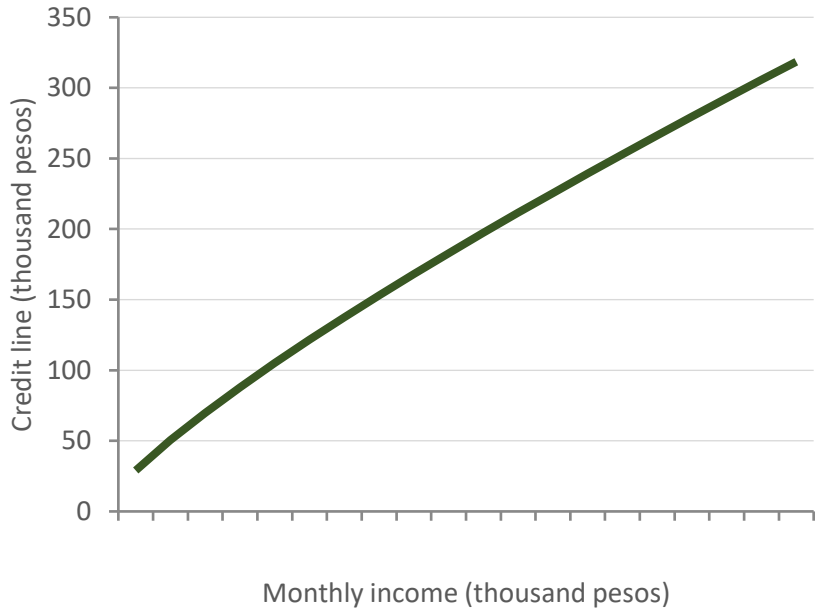
EXAMPLE (ILLUSTRATIVE VALUES)

β = 0.8: +1% income goes with about +0.8% credit line. And +10% income? 1.10^0.8 − 1 = +7.9%.

WHERE IT COMES FROM

$\ln(y_1) - \ln(y_0) = \beta \cdot \ln(1.01) \rightarrow y_1 / y_0 = 1.01^{\beta} \approx 1 + 0.01 \cdot \beta$

Shape: a curve (here β < 1)



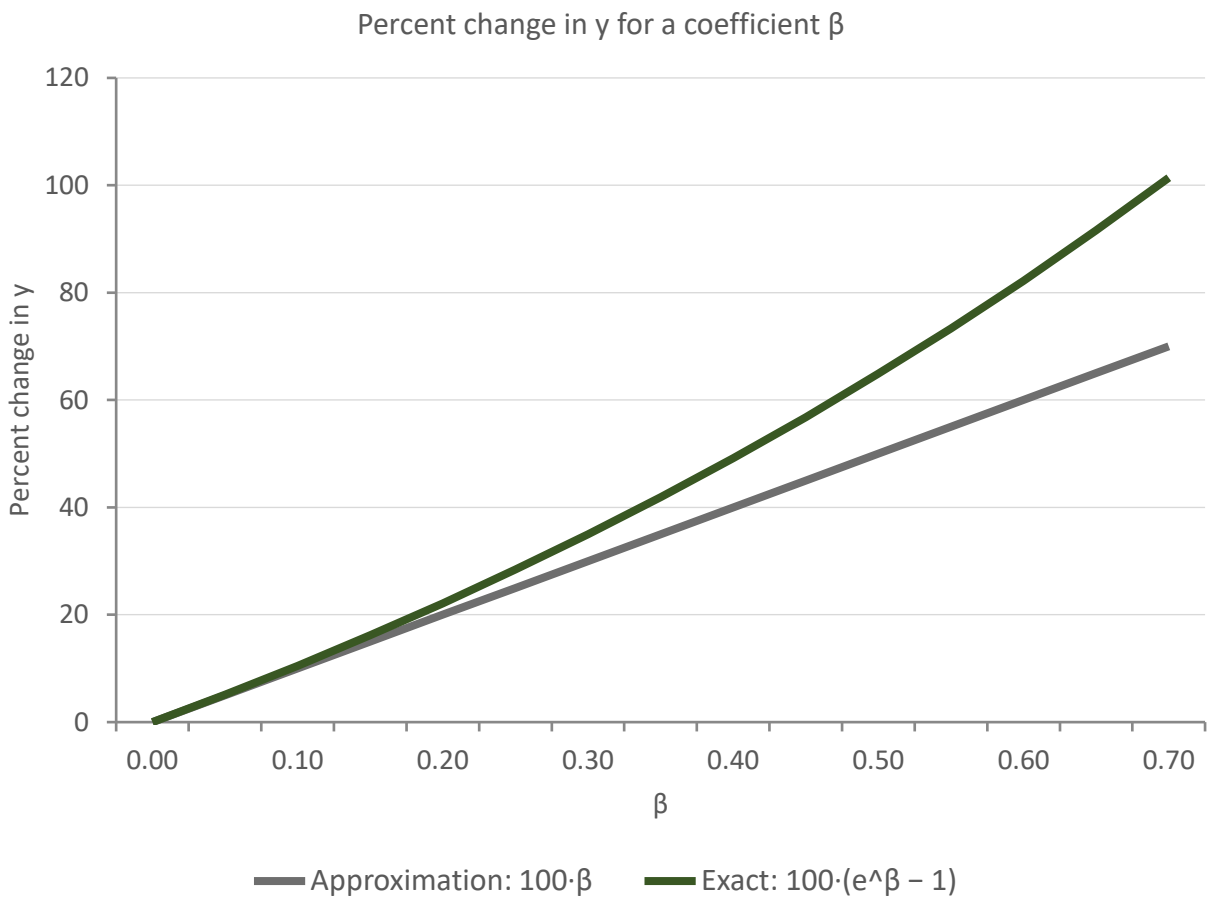
Needs x > 0 and y > 0. It also tames skewness in x, so it is the usual first try for money variables.

Four models, one reading rule

Model	Equation	x changes by	Effect on y
LIN–LIN	$y = \alpha + \beta \cdot x$	+1 unit	y changes by β units
LIN–LOG	$y = \alpha + \beta \cdot \ln(x)$	+1%	y changes by β / 100 units
LOG–LIN	$\ln(y) = \alpha + \beta \cdot x$	+1 unit	y changes by ≈ 100·β % (exact: 100·(e^β − 1) %)
LOG–LOG	$\ln(y) = \alpha + \beta \cdot \ln(x)$	+1%	y changes by ≈ β % (exact: 100·(1.01^β − 1) %)

Whoever carries the log is read in percent.

When β as a percentage stops working



β = 0.03

Approx. +3.0% · Exact +3.05%

β = 0.085

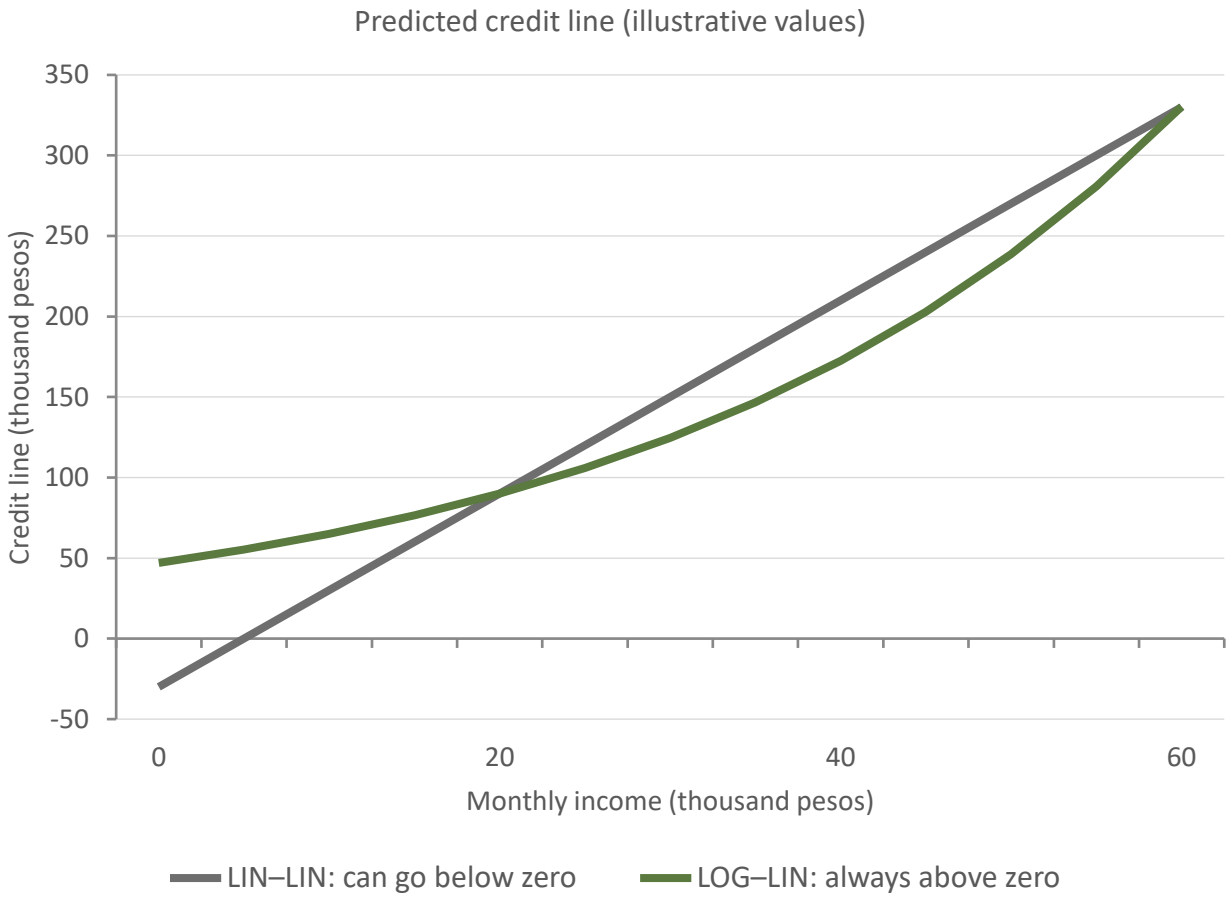
Approx. +8.5% · Exact +8.9%

β = 0.5

Approx. +50% · Exact +64.9%

If |β| < 0.1, read β as %. Otherwise report e^β − 1.

Log on y keeps predictions positive



exp(anything) > 0

The model predicts ln(ŷ). Back in pesos, ŷ = exp(·), which can never be negative.

Positive, not accurate. It cannot go below zero, but it can still miss.

A bonus, not the main reason. The main reason is proportional effects.

Needs y > 0 to begin with. With real zeros or negatives, the log does not exist.

Back to pesos: exp() gives the median, not the mean

The model predicts

$\ln(\hat{y}) = 11.60$

→

Apply exp()

≈ 109,100 pesos

→

What you get

≈ the median of y for that client

Always back-transform before computing MAE, MAPE or RMSE: errors must be in pesos.

Does this variable deserve a log?

LOG ON y

y is strictly positive and effects are proportional: a richer client gets a percentage more credit, not a fixed amount.

LOG ON x

x is positive and right-skewed, and returns diminish: the 100,000th peso of income adds less than the 10,000th.

Keep it in levels when

- There are zeros or negatives you cannot justify
- x is a dummy variable
- x is bounded or symmetric, like score_buro
- One unit already means something, like a year of age

Decide variable by variable: $\ln(\text{monto}) \sim \ln(\text{ingreso}) + \text{edad} + \text{score_buro}$

What do the diagnostics say?

Symptom	Where you see it	Try
The relationship is curved	RESET rejects; residuals vs. fitted show a U shape	**Log on x (LIN–LOG)**
Residuals fan out	Residuals vs. fitted: the spread grows with the fitted value	**Log on y (LOG–LIN)**
Residuals are right-skewed	Q-Q plot: the right tail bends away from the line	**Log on y (LOG–LIN)**
Curved and fanning out	Both plots show a problem	**Log on both (LOG–LOG)**
Straight and even	All the plots look clean	**No log needed (LIN–LIN)**

These are rules of thumb. After transforming, run RESET and the residual plots again: the fix has to show up.

Compare models on the same scale

1 Look at the shape

Scatter y vs x, and the same plot with ln on one or both axes.

2 Check the diagnostics

Residuals vs fitted, Q-Q plot, and the RESET test.

3 Compare in pesos

Back-transform predictions, then MAE, MAPE and RMSE on the test set.

The R^2 trap

R^2 of a LIN model measures variation in y. R^2 of a LOG model measures variation in ln(y). They are not comparable.

QUICK GUIDE

y positive and right-skewed: log y

Diminishing returns in x: log x

A “% per %” question: log–log

YOUR TURN · 5 MINUTES

Say it in one sentence

LIN–LIN $\beta = 1{,}200$

Credit line (pesos) vs. client age (years)

How would you explain this coefficient to the risk team?

LOG–LIN $\beta = 0.004$

ln(credit line) vs. months at current job

How would you explain this coefficient to the risk team?

LIN–LOG $\beta = 45{,}000$

Credit line (pesos) vs. ln(monthly income)

How would you explain this coefficient to the risk team?

LOG–LOG $\beta = 0.6$

ln(credit line) vs. ln(monthly income)

How would you explain this coefficient to the risk team?

Check your sentences

LIN–LIN $\beta = 1{,}200$

Credit line (pesos) vs. client age (years)

Each extra year of age goes with 1,200 pesos more credit line.

LOG–LIN $\beta = 0.004$

ln(credit line) vs. months at current job

Each extra month goes with about +0.4% credit line. Each extra year (12 months): e^0.048 − 1 ≈ +4.9%.

LIN–LOG $\beta = 45{,}000$

Credit line (pesos) vs. ln(monthly income)

1% more income goes with 450 pesos more credit line (45,000 / 100).

LOG–LOG $\beta = 0.6$

ln(credit line) vs. ln(monthly income)

1% more income goes with about +0.6% credit line.